28th SPVM National Physics Conference (SPVM 2026)

A Practical System for Optimizing Large-Scale Conference Programs

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01

Background and Objective

Survey / Previous Studies / Objective

What Is an Academic Conference?



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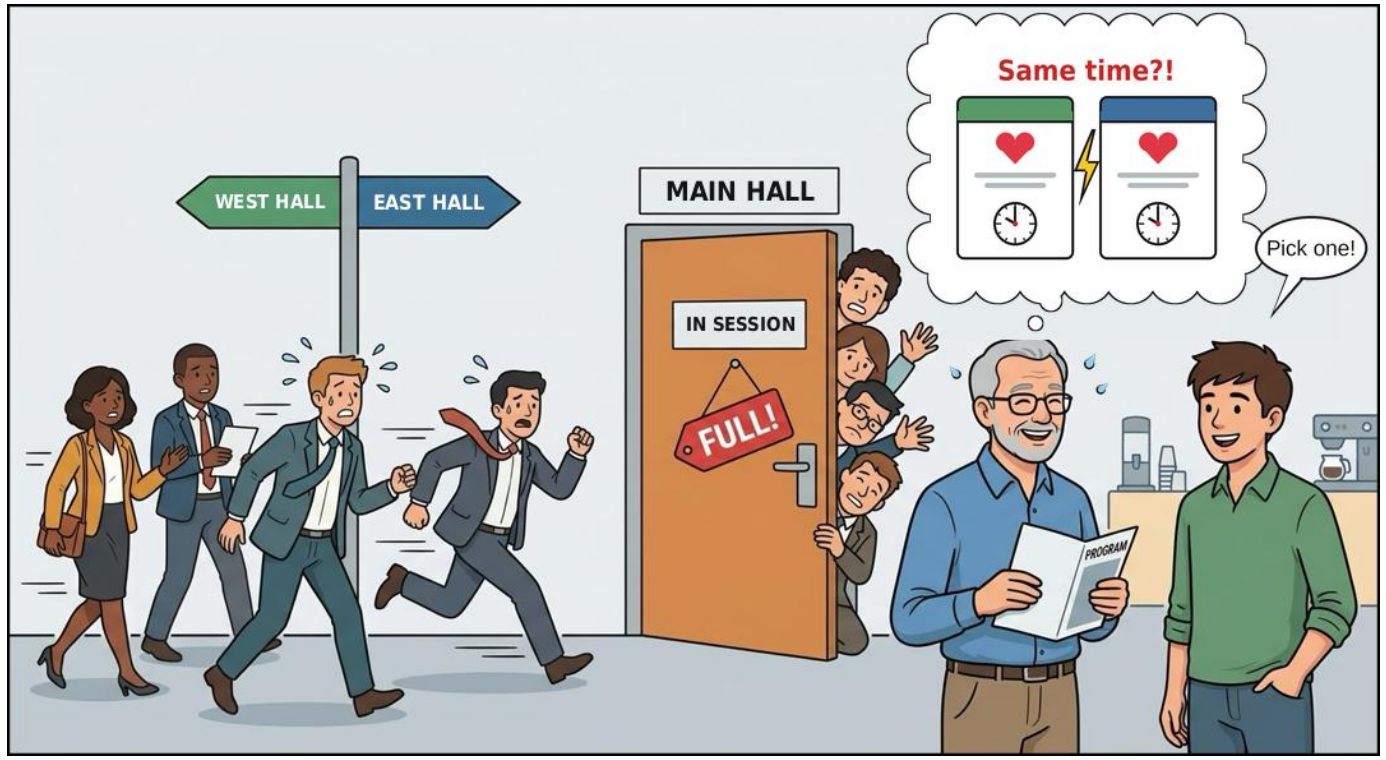
At a conference

1. **Oral presentations**
2. **Poster presentations**
3. **Networking among researchers**

▼

Advancing science

The Problem: Making a Good Program Is Hard



Generated by Gemini

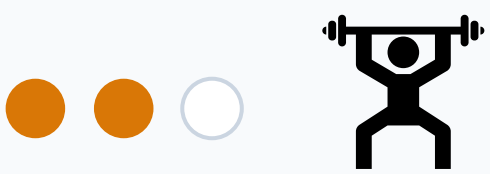
A bad program causes...

- **A** Moving between rooms
- **B** Two talks you want, at the same time
- **C** Full rooms (or empty rooms)

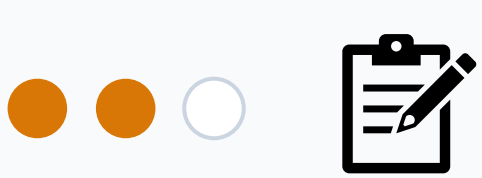
A good program avoids A, B and C

Survey: Administration of the Academic Conference

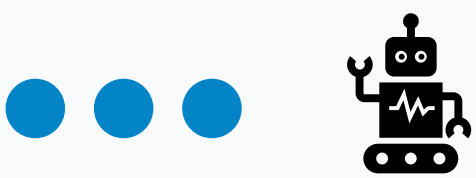
Scheduling is “a lot of work”



Done by hand in spreadsheets



Open to automated tools



A lot of work, done by hand — and tools are welcome

All answers → Appendix A2

Previous Studies: Powerful, but Hard to Use

Previous studies

[1] 3 problems solved in a row

$$\text{Phase 1} \quad \min \sum_{e \in H} c_e x_e$$

$$\text{Phase 2} \quad \min \sum_{b} w_b y_b$$

$$\text{Phase 3} \quad \min \sum_{b} \sum_{t} u_{b,t} z_{b,t}$$

[2] 6 weights × 6 penalty tables

$$\min \sum_{s} \sum_{r} \sum_{t} \left(w_\alpha \alpha_s^t + w_\beta \beta_r^t + w_\gamma \gamma_{s,r} \right) Z_{s,r}^t$$

$$+ \sum_{s} \sum_{r} \sum_{(t,su)} \left(w_\delta \delta_s^{(t,su)} + w_\varepsilon \varepsilon_s^{(t,su)} + w_\zeta \zeta_r^{(t,su)} \right) X_{s,r}^{(t,su)}$$

Input: every participant's preferences [1] / 9 sheets + 6 weights [2]

This study

$$100\left(w_1 \frac{A}{N_A} - w_2 \frac{B}{N_B} - w_3 \frac{C}{N_C} \right)$$

A continuity B conflicts C empty seats

Input: 3 sheets + 3 weights

[1] Vangerven et al. 2018, Eqs. (1)(4)(8) [2] Pylyavskyy et al. 2024, Eq. (11) Details → Appendix A13

Our Goal

Creating a “Simple Conference Scheduler”

What we did

1. **Build the system**
2. **Check that it works**
3. **Evaluate with a real conference**



We want to make running a conference easier.

02

System Design

Mathematical Optimization / Formulation / Using the App

Updated Since the Abstract

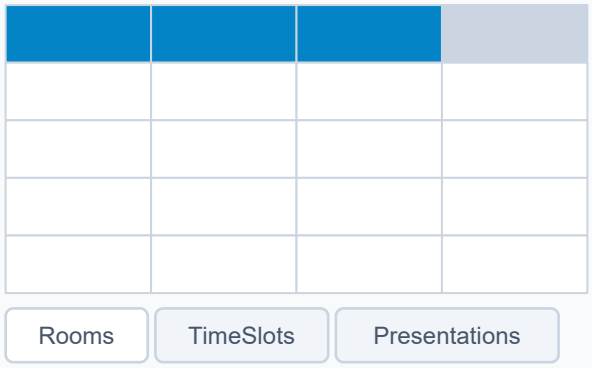
The system has been updated since the Abstract → Today we present the new version

	Today
B Topic conflicts	Counted
Score	Each term ÷ N (same scale)
Time (40 talks, equal weights)	≈ 1,400 s

Details → Appendix A3

System Overview

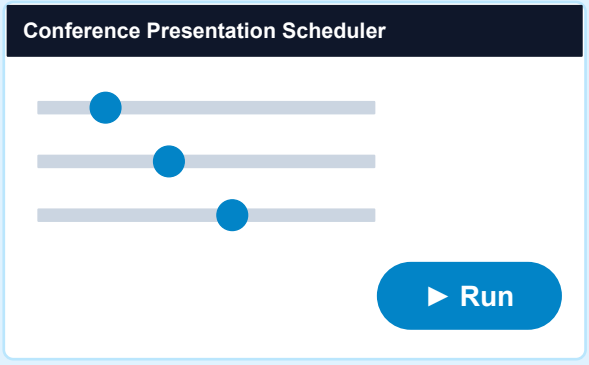
① Input Excel



Rooms | TimeSlots | Presentations

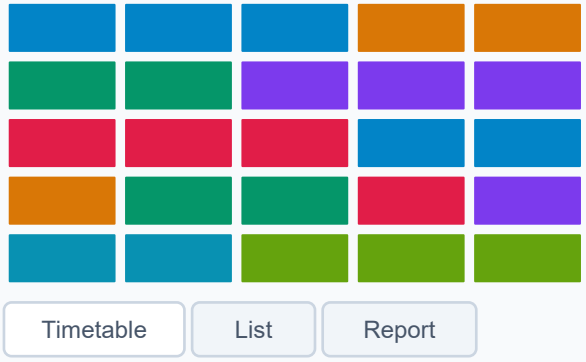
3 sheets · 7 columns

② Compute App



1 click

③ Output Excel

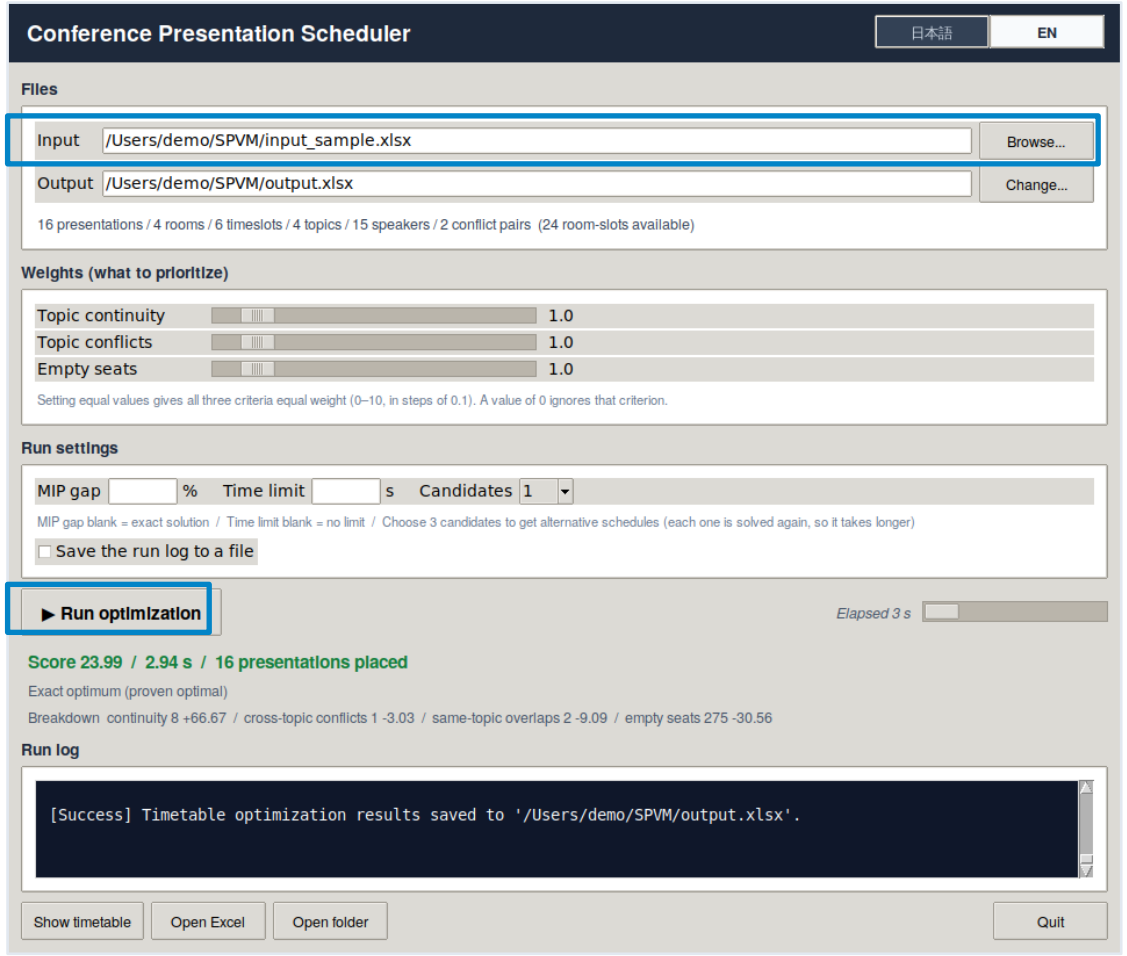


Timetable | List | Report

Timetable · List · Report

Prepare Excel → Press Run → Get the timetable

App Screen: How to Use



Required **Required steps**

1. **Choose the input file**
2. **“Run optimization”**

Optional **Optional (defaults work)**

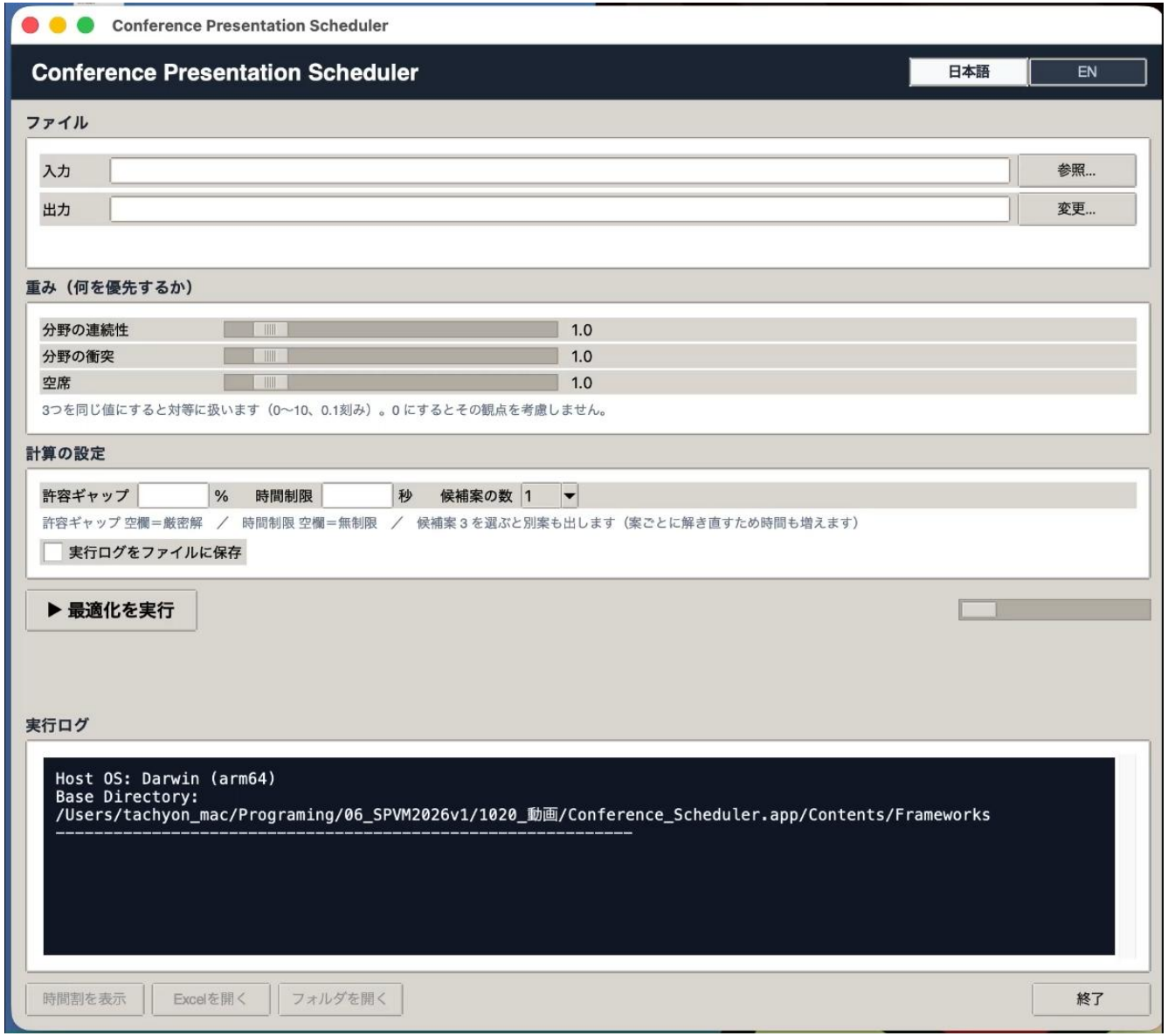
3. **Weights (0–10, default 1.0)**
4. **Gap · time limit · candidates**
5. **Save the run log**

Result **Check the result**

6. **Score and breakdown**

Settings → Appendix A4

App Screen: How It Works

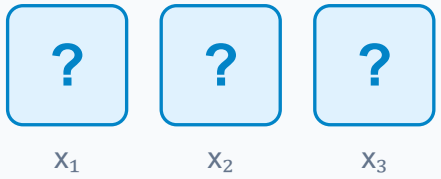


What Is Mathematical Optimization?

Find the best choice that follows the rules

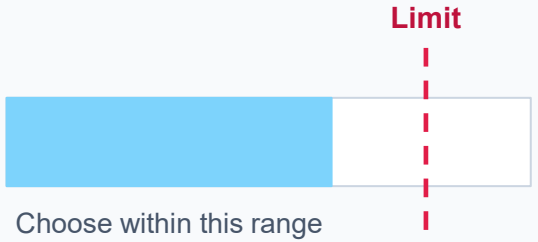
Variables

What we decide



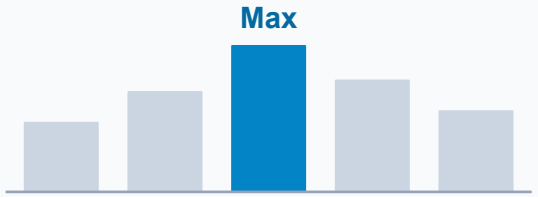
Constraints

Rules to follow



Objective function

A score for “how good”



Compared with AI

- ✓ **Always follows the rules**
- ✓ **Reasoning is transparent**
- ✓ **Scores are reproducible**

Formulation ①: Variables and Constraints

Variable: is this talk here? (1 = yes, 0 = no)

	Time 1	Time 2	Time 3	Time 4	Time 5
Room A	0	0	0	0	0
Room B	0	0	**1**	0	0
Room C	0	0	0	0	0
Room D	0	0	0	0	0

Talk P01 → Room B, Time 3

Always

1. **Each talk: exactly one place**
2. **Each room and slot: at most one talk**
3. **Audience ≤ room capacity**

Only if the column is filled

- **Screen**
- **Same speaker**
- **Unavailable slot**

More → Appendix A6

Formulation ②: Objective Function

Sco

A

Same topic con
same room

+1

Room

Slot →

The audience nee

Minimize

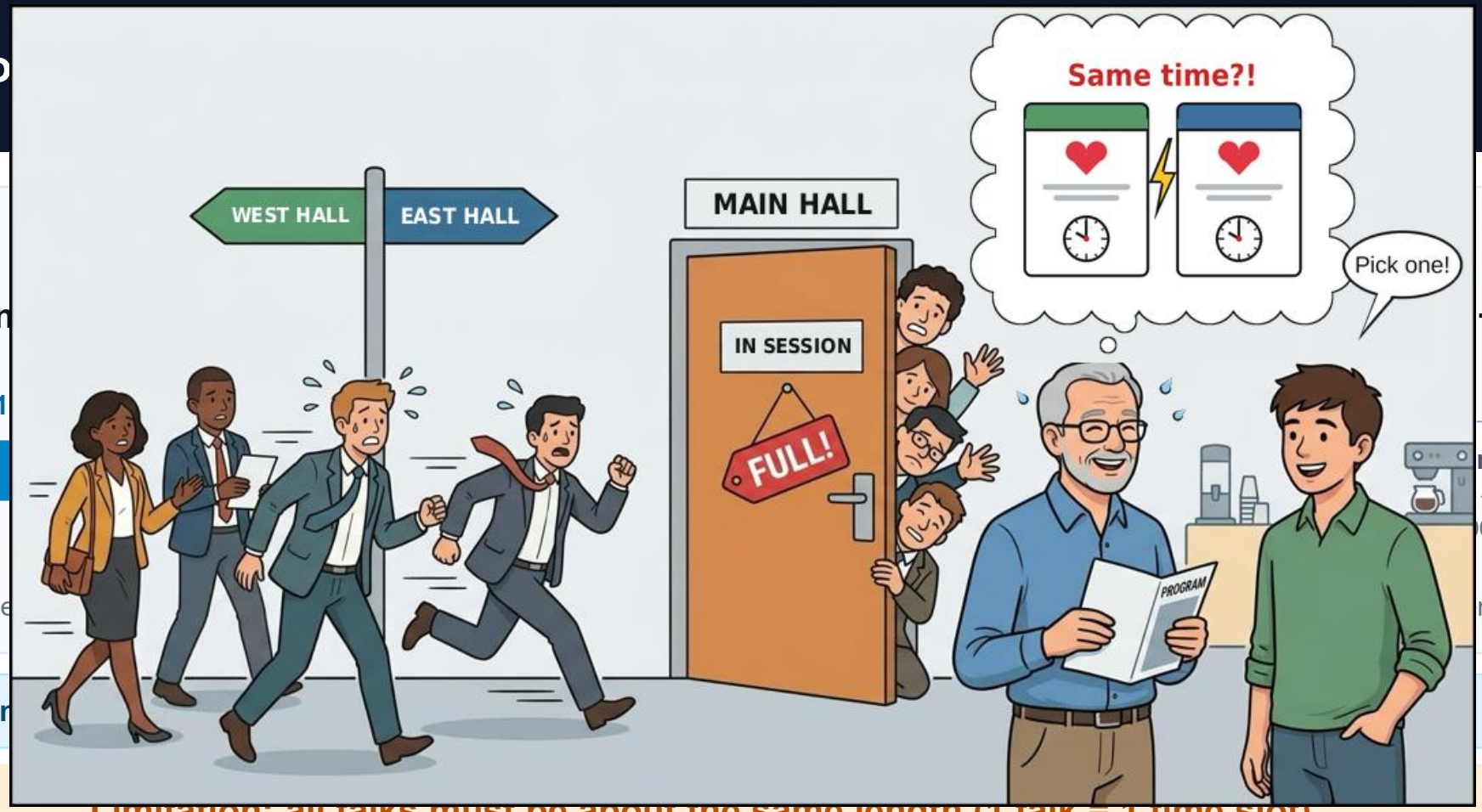
–

mpty 40

0

nce size

w: how importar



Limitation: all talks must be about the same length (1 talk = 1 time slot)

More → Appendix A7

03

Results and Discussion

Testing the Weights / Data Modeled on a Real Conference

Two Tests

Test 1: Does it work?

Same data as the Abstract

40 talks	**10** rooms
5 slots	**7** topics



Test 2: Is it practical?

Based on a real conference

89 talks	**4** rooms
24 slots	**8** topics

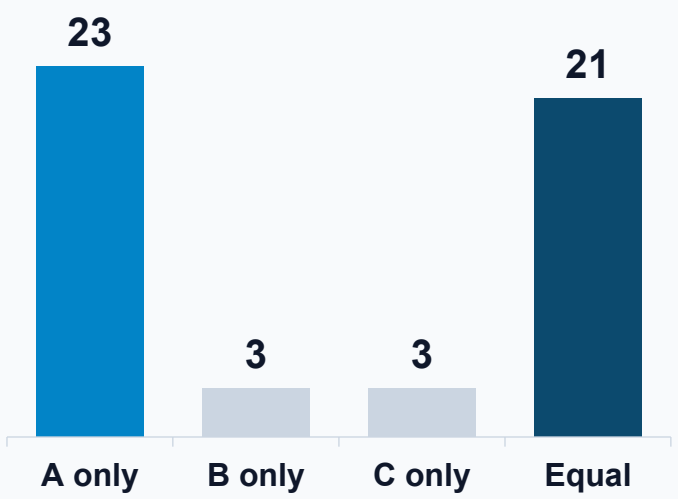


All runs: my laptop — MacBook Air (M4, 16 GB)

Test 1: The Weights Work as Intended

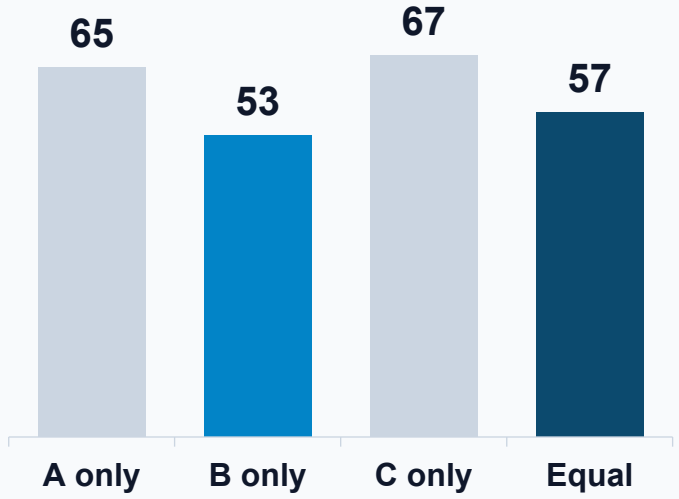
A Topic continuity

higher is better



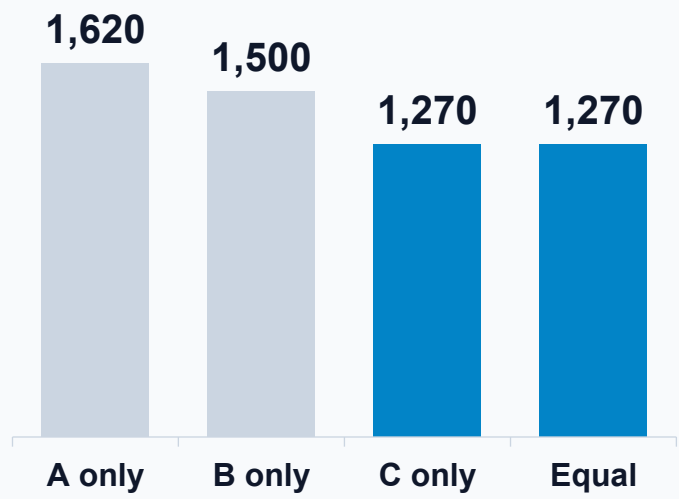
B Topic conflicts (penalty)

lower is better



C Empty seats

lower is better



best for that point

equal weights (1 : 1 : 1)

Each weight improves its own point — equal weights are near the best on all three

Input data and timetables → Appendix A8, A9

Test 2: Input data and conditions

Input: based on real physics conference

89	4	24	8
Talks	Rooms	Time slots	Topics

Settings: equal weights (1 : 1 : 1), time limit 90 s

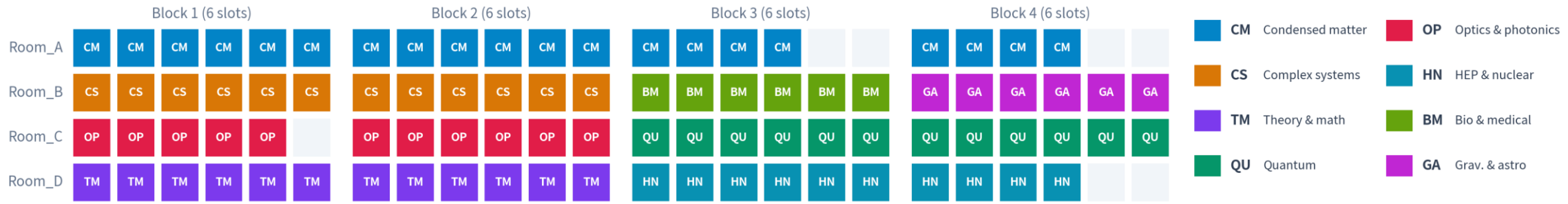
Data details → Appendix A10



Public data is for imitation;
unknown data is for inference.

Test 2: Demonstrated its practicality

Real program



This system (90 s)



A Topic continuity

73 → 72 −1%

B Topic conflicts

51 → 39 −24%

C Empty seats

1,640 → 1,640 ±0%

Time

90 s

Similar to the real program → Practical

04

Conclusion

Summary / Released on GitHub

Summary

1	Built a system that makes a first draft of a conference program with mathematical optimization	3 sheets · 7 columns · 1 click
2	Confirmed that it works with test data	weights work: 4 / 4 settings
3	Showed that it is practical, using a real conference as a model	A −1% · B −24% · C ±0% · 90 s
4	Weak point: only for conferences where talks are about the same length	1 talk = 1 time slot

Thank You for Your Attention

References

[1] B. Vangerven et al. Omega. 38 (81) 2018

[2] Y. Pylyavskyy et al. Eur. J. Oper. Res. 487 (317) 2024

Appendix (A1–A13)

A1	Roadmap	A8	Test 1: input data
A2	Survey answers	A9	Test 1: results
A3	Changes since the Abstract	A10	Test 2: data
A4	App settings	A11	Test 2: results
A5	Optimization (more)	A12	PuLP / CBC / HiGHS
A6	Variables & constraints	A13	Previous studies
A7	Objective function		

These slides and the code are also on GitHub

GitHub

https://github.com/Tachyon327/Conference-Scheduler

A

Appendix

Supplementary material for Q&A

APPENDIX

Appendix Contents

One topic per page. During Q&A, jump to a page by its number.

Supplements to the talk

A1 Roadmap
Planned updates until March 2027

A2 Survey answers
All answers from 3 conferences

A3 Changes since the Abstract
What was fixed, and why time grew

A13 Previous studies
Input, time, solver

How the system works

A4 App settings
Weights, gap, time limit, candidates

A5 Optimization (more)
MILP and branch-and-bound

A6 Variables & constraints
All rules, with formulas

A7 Objective function
A, B, C and the scale N

Data and results

A8 Test 1: input data
40 talks / 10 rooms / 5 slots

A9 Test 1: results
Timetables and numbers

A10 Test 2: data
What is real, what is assumed

A11 Test 2: results
Both timetables, A / B / C

Solver

A12 PuLP / CBC / HiGHS
Why another solver may be faster

APPENDIX A1

Roadmap (updates planned until around March 2027)

1	Switch the solver from CBC to HiGHS	Become faster
2	Improve the definition of objective B (conflicts between simultaneous presentations)	More useful
3	Build a Windows app and support more macOS versions	Delivering to everyone
4	Scheduled for maintenance and improvements through March 2027	Until My Graduation

APPENDIX A2

Survey Answers (3 Conferences, Anonymous)

Three interviews. All three conferences agreed to publication if anonymous.

Question	Conference A	Conference B	Conference C
Respondent's role	Program / organizing committee (researcher, faculty)	Conference secretariat staff	Program / organizing committee (researcher, faculty)
How the program is made	Manual work and adjustment in spreadsheets	Schedule in spreadsheets; program in an in-house submission and program-editing system	Manual work in spreadsheets (Excel, Google Sheets)
External vendor tools	—	In-house program and optimization tool	None (done in-house with Google Forms)
Scheduling burden	Takes considerable effort	Not much effort	Takes considerable effort
Main difficulties / focus	Handling incomplete submissions and individual requests	Avoiding speaker double-booking / individual availability and equipment / avoiding conflicts between similar topics / room capacity (standing, empty seats)	Topic continuity (same topic in the same room) / last-minute cancellations and changes
Interest in using a tool	Would consider it, depending on conditions (usability, cost, etc.)	Would consider it, depending on conditions (usability, cost, etc.)	Would definitely like to try
Free comments	—	—	Choosing which accepted submissions beyond the oral slots move to posters, and notifying authors, takes effort (summary)

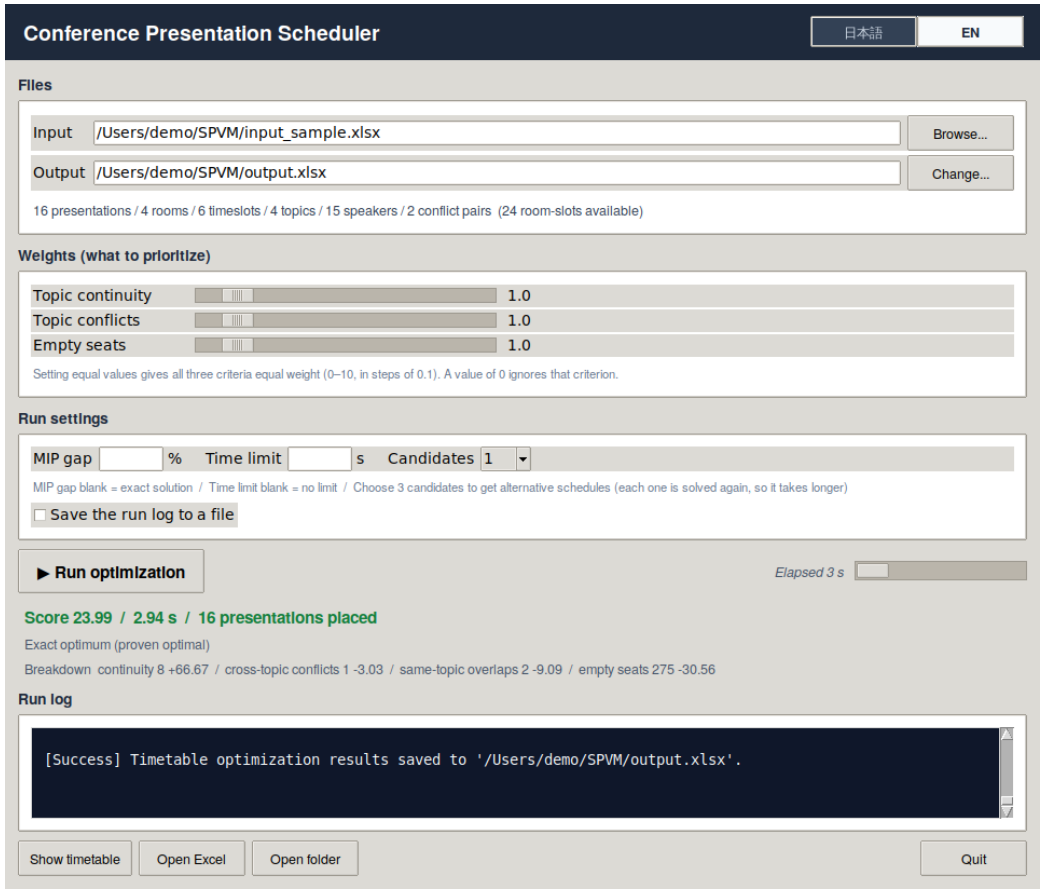
APPENDIX A3

Changes Since the Abstract

	Abstract	Now
Topic conflicts	Penalty was always 0 (bug)	Fixed; split into B1 (similar topics) and B2 (same topic)
Score	W1·A − W2·B − W3·C	Each part ÷ its possible range, × 100
Computation time	≈ 566 s / 7 s / 0.1 s	Longer: 1,360–1,455 s (equal weights, 3 runs)
Interface	Excel only	Desktop app (weights, MIP gap, time limit, candidates)
New rule	—	Same speaker not in two rooms at once

APPENDIX A4

App Settings



Setting	Range / default	What it does
Input / output file	required	Excel in, Excel out
Weights A, B, C	0–10 (step 0.1), default 1.0	Which point matters more. 0 = ignore it
MIP gap	%, blank = 0 (exact)	Stop when the score is this close to the proven best
Time limit	seconds, blank = none	Stop and return the best timetable so far
Candidates	1–3, default 1	Alternative timetables (each is solved again)
Run log	on / off	Save the solver log to a file
Result panel	—	Score, breakdown (A, B1, B2, C), quality: proven / time limit

APPENDIX A5

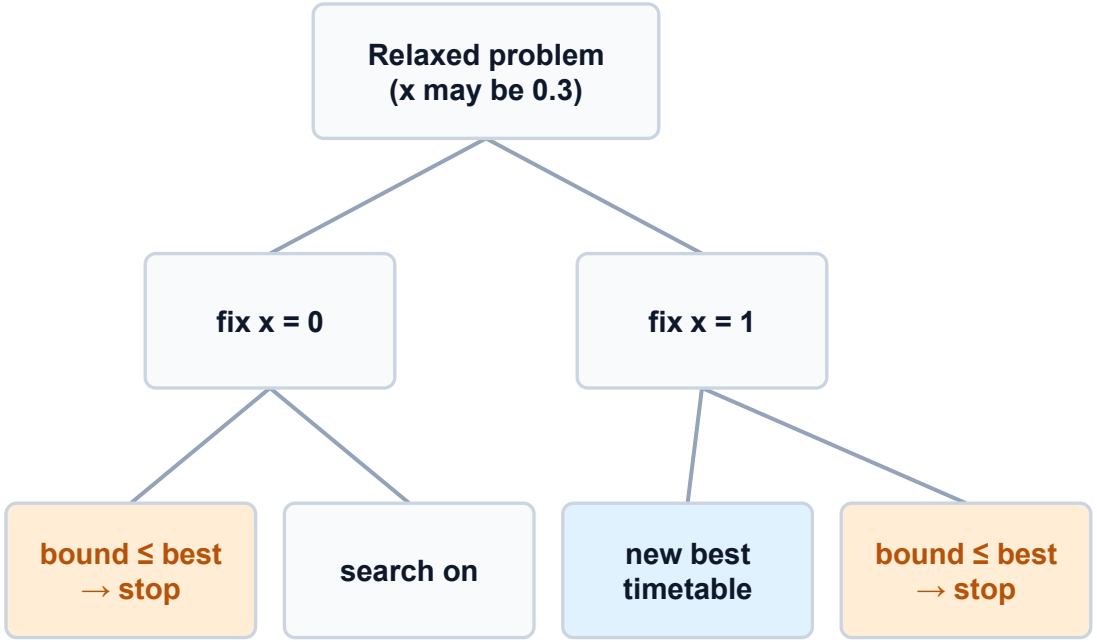
Mathematical Optimization in More Detail

Variables	x = 0 or 1 → integer
Constraints	sums of x ≤, = numbers → linear
Objective	sum of x with weights → linear

Mixed-Integer Linear Programming (MILP)

Solver: CBC (free) through PuLP (Python).
Exact run: the solver proves that no better timetable exists (gap 0).
Time-limited run: returns the best timetable found so far (not proven).

Branch-and-bound (how the solver searches)



Branches that cannot beat the best timetable are cut. Proving “no better one exists” is often the slow part.

APPENDIX A6

Variables and Constraints in Detail

x(s, r, t) = 1 if presentation s is in room r at time slot t, otherwise 0

Rule	Formula	When
Each presentation exactly once	$\Sigma_r \Sigma_t x(s, r, t) = 1$ for every s	Always
At most one presentation per room and slot	$\Sigma_s x(s, r, t) \leq 1$ for every r, t	Always
Room large enough	x(s, r, t) = 0 if capacity(r) < audience(s)	Always
Screen	x(s, r, t) = 0 if s needs a screen and r has none	If Needs_Screen is filled
Unavailable slot	x(s, r, t) = 0 for t in NG(s)	If NG_TimeSlot_ID is filled
Same speaker	$\Sigma_{s \text{ of speaker } p} \Sigma_r x(s, r, t) \leq 1$ for every p, t	If Speaker is filled
Same length	1 presentation = 1 time slot	Model assumption

Variables that break an “x = 0” rule are not created at all, which keeps the model small.

APPENDIX A7

Objective Function in Detail

$$\max 100\left(w_1 \frac{A}{N_A} - w_2 \frac{B}{N_B} - w_3 \frac{C}{N_C}\right)$$

Term	What is counted	Scale N (the range it can move)	N: Test 1 / Test 2
A continuity ↑	+1 when the same topic is in the same room in two neighbouring slots (same day)	N_A = talks − topics	33 / 81
B1 cross-topic ↓	+Level (1–3) for each pair of similar topics in the same slot	N_B = slots × ΣLevel + 3 × (talks − topics)	139 / 651
B2 same topic ↓	+3 for each extra talk of the same topic in the same slot	(included in N_B)	—
C empty seats ↓	capacity − expected audience, summed over all talks	N_C = Σ (largest − smallest usable room)	4,910 / 2,760

Why divide by N: the raw sizes differ (empty seats ≈ 1,000, continuity ≈ 20). After ÷N each term is 0–1, so w1 : w2 : w3 means what it says.

Known issue: B1 counts whether two topics meet in a slot, not how many talks meet.
With weight only on B, one topic can be packed into one slot (Test 1). A fix is planned.

APPENDIX A8

Test 1: Input Data (Same as the Abstract)

40	10	5	7
talks	rooms	slots (Day 1: 3, Day 2: 2)	topics

Room	Capacity	Screen
R01	200	yes
R02, R03	150	yes
R04	100	yes
R05, R06	80	yes
R07	40	yes
R08	40	no
R09, R10	30	no

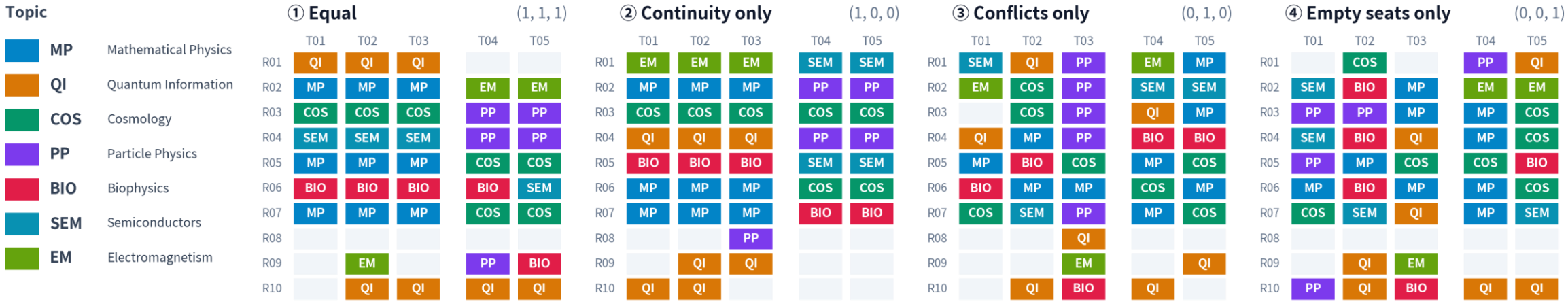
Topic	Talks
Mathematical Physics	9
Cosmology	7
Quantum Information	7
Biophysics	5
Particle Physics	5
Semiconductors	4
Electromagnetism	3

Similar topics (Level): Particle–Cosmology 3, Particle–Math. Phys. 2, Bio–Quantum Info. 2, Semicond.–EM 1
Audience 20–150 people per talk · 20 talks need a screen · 3 talks have an unavailable slot (P32: T05, P33: T01, P36: T01)

APPENDIX A9

Test 1: Detailed Results

Exact solutions (gap 0, no time limit) · MacBook Air M4 · bundled CBC 2.10.13 · 3 runs each, same timetable every run



Weights (A, B, C)	Score	A continuity	B1 pairs	B2 overlaps	B penalty	C empty seats	Time (s), 3 runs
① **Equal (1, 1, 1)**	−3.24	21	9	13	57	1,270	1,455 / 1,425 / 1,360
② **A only (1, 0, 0)**	69.70	23	7	16	65	1,620	148 / 157 / 161
③ **B only (0, 1, 0)**	−38.13	3	9	12	53	1,500	0.21 × 3
④ **C only (0, 0, 1)**	−25.87	3	10	15	67	1,270	0.02 × 3

B penalty = Σ Level (B1) + 3 × B2. The score of each run uses its own weights, so scores are not compared across rows.

APPENDIX A10

Test 2: Data Based on a Real Conference

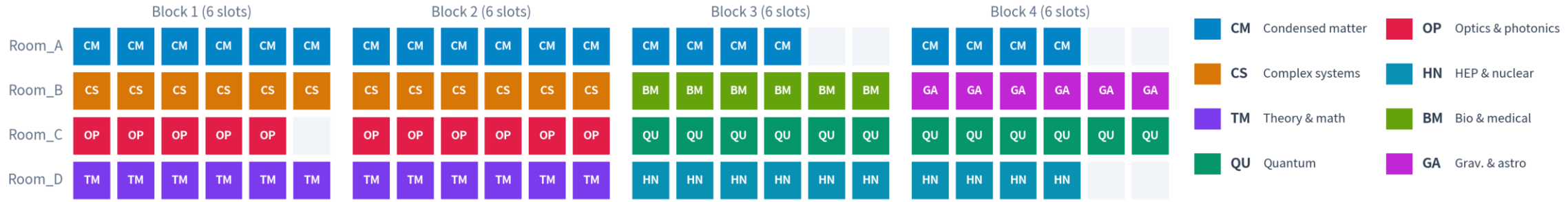
Based on the public program of a physics conference (name, dates and venue withheld)

Item	Content	Grounds
Presentations	89 parallel oral presentations (16 sessions)	Excluded: special sessions incl. invited talks (different length, separate venue), plenaries, posters
Rooms / slots	4 rooms, 20 min × 6 slots × 4 blocks = 24 slots	As in the public program (times also as actual)
Topics	8 topics (max 20, min 6 talks)	Topics taken from session names. No one speaks twice (2 pairs sharing a surname confirmed to be different people)
Capacity	Room_A 100, Room_B–D 60 each	No public capacity data. Room_A always hosts the largest topic and is used for plenary events, so it is assumed to be larger
Expected attendees	Proportional to topic size (largest topic, 20 talks = 80 people)	No audience data. Consequence: the largest topic fits only in Room_A (as in reality)
Conflicting topic pairs	9 pairs (Level 1–3)	Set by topic similarity (e.g., high energy – gravitation/astro 3, quantum – optics 2)
Blocks / equipment	Continuity across blocks is not counted. Screens and unavailable slots are not used	In reality, room topics change between morning and afternoon. All rooms assumed to have screens

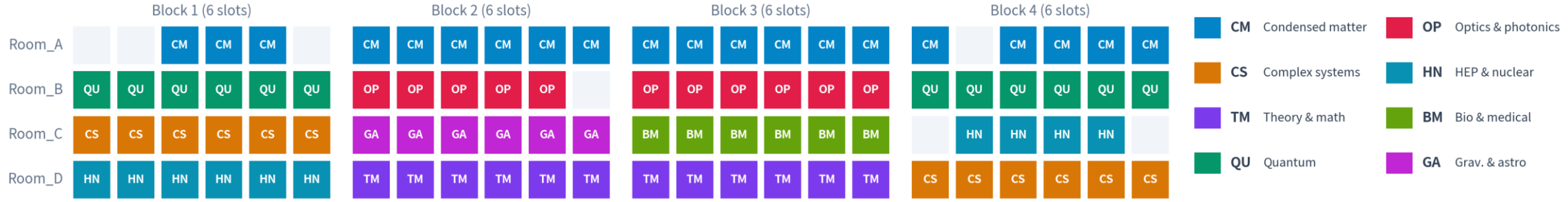
APPENDIX A11

Test 2: Detailed Results

Real program



This system: equal weights, time limit 90 s



	A continuity	B1 pairs	B penalty	B2 overlaps	C empty seats	Score (1 : 1 : 1)	Time
Real program	73	35	51	0	1,640	22.87	—
This system	72	25	39	0	1,640	23.48	88.8 s (limit 90 s)
Difference	−1.4%	−29%	−24%	±0	±0%	+0.61	

APPENDIX A12

PuLP / CBC, and Why HiGHS May Be Faster

PuLP

Python library.
We write the variables,
rules and score here.

CBC

Free solver bundled with PuLP.
Used in this study.
No symmetry detection by default.

HiGHS

Newer free solver (MIT).
Detects symmetry
(e.g. identical rooms) automatically.

Test 1 weights	CBC (MacBook Air, s)	HiGHS (other PC, s)	Same optimum?
(1, 1, 1)	1,360–1,455	227	yes (−3.24)
(1, 0, 0)	148–161	8.5	yes (69.70)
(0, 1, 0)	0.21	4.2	yes (−38.13)

- Second run of (1, 1, 1) with HiGHS: 254 s with symmetry detection, 832 s without → one reason, not the only one.
- Public benchmark (Mittelmann, MIPLIB 2017, 240 problems, Oct 2025): HiGHS solved 157, CBC 107.
- PuLP can call HiGHS (pip install highspy). Not yet built into the app.

Different machines: this is not a strict comparison. To be re-measured on the MacBook Air.

APPENDIX A13

Detailed Comparison with Previous Studies

Aspect	[1] Vangerven et al. (2018) Omega 81, 38–47	[2] Pylyavskyy et al. (2024) EJOR 317, 487–499	This study
Aim / strengths	Maximizes attendance from participants' session preferences, then also considers room hopping and presenter availability step by step (3-phase hierarchical optimization). Used at 4 real conferences	A system of penalties and weights handles in-person, hybrid and online formats, multi-slot presentations, topic–room matching, keeping similar topics apart, etc. in one model	Make a first draft from minimal input, with free software
Input	Session preferences of all participants (e.g., ORBEL 2017: 1,200 in total from 101 of 140 participants) and presenter availability	9 sheets (Submissions, Tracks, Sessions, Rooms, Parameters, penalty tables, Similar Tracks) and 6 weights	3 sheets, 7 required columns + 3 weights
Computation time	0.03–61 s per phase; but the k-block search (arrangements with less room hopping) takes hours	Time limit 1 h. Most runs finish within minutes (exact model: 0.1–137 s except GECCO19), but GECCO19 and 3 cases of the extended model hit the limit. OR60F3 takes 1,072.7 s just to build the model	40 talks: 1,360–1,455 s (exact, equal weights) · 89 talks: 90 s limit
Solver / environment	CPLEX 12.6.3 (commercial), laptop (Core i7, 8 GB)	Gurobi 9.5.0 (commercial) + Python 3.8.3, Core i7, 16 GB	CBC (free) via PuLP, MacBook Air M4 16 GB
Relation to this study	Same assumption as this study: all presentations have the same length	Handles multi-slot presentations, topic–room matching, etc. This study omits these to keep the input minimal	All presentations have the same length (1 slot)